

An AI-Powered Simulation and Tutoring Application for Evolutionary Psychology

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Abstract

We present an interactive web application designed to teach core concepts in evolutionary psychology through agent-based simulations and an AI-powered Socratic tutor. The system targets undergraduate students who find abstract topics such as self-domestication, theory of mind, and cultural transmission difficult to grasp. Users manipulate simulation parameters, observe behavioral dynamics, and engage a conversational tutor that withholds answers, probes reasoning, and poses a transfer question once understanding is demonstrated. We describe the learning-theoretic grounding, system architecture, a four-version prompt optimization process evaluated against a 17-scenario rubric (final score 66/72), findings from six user-testing sessions, an ethical analysis, and a productionization plan with cost modeling and data-privacy considerations.

Keywords

evolutionary psychology, AI tutoring, interactive simulation, transfer learning, prompt optimization, educational technology

1 Introduction & Problem Statement

When people think about “evolution,” they would likely think about how genes for physical attributes get passed down from generation to generation, leading to changes observed between different eras. However, evolution is a more nuanced topic which the Evolutionary Psychology class at Occidental College looks into, such as by discussing how culture or social behavior led to the modern species seen today. When asking Professor Silver, Occidental College’s evolutionary psychology professor, he mentioned how some concepts in this class can be difficult for students to understand. As such, he makes use of weekly quizzes to make sure students understand the current topics, often by having students think critically about how the material connects to what was previously discussed. One key example of this is when students were asked to look at the behavior of elephants, and determine whether elephants were a self domesticated species based on how other species, such as chimpanzees and bonobos, behaved. In this instance, self domestication means a species undergoing behavioral, morphological, physiological, and cognitive changes associated with decreased aggression in the absence of intentional breeding by humans. However, Professor Silver mentioned how some concepts are more difficult for students to grasp, including culture and theory of mind. Given this fact, we created an application primarily targeted towards current (or

incoming) evolutionary psychology students to teach them about recurring topics (self domestication/aggression) and topics which students typically have trouble with. Users will interact with various simulations, playing around with them to understand the current topic, and having the opportunity to ask an AI tutor any questions. After the tutor has determined the student demonstrates a clear understanding of the topic, it will prompt the student to think about it in the context of a different species, evaluating and providing feedback based on the user’s answer. Given how nuanced and difficult evolutionary psychology might be, having a tutor powered by AI would allow for dynamic responses that best suit the user, similar to that of going to a tutor or professor. Without using AI, the user themselves would be responsible for determining if they have gained an understanding of the material, which may be subject to their own bias. This may be because they take the information given to them at face value, which may lead to an incorrect understanding. Additionally, they would not have the ability to ask clarifying questions that would help in learning the material in a timely manner as tutors and professors are not available at every time of day.

2 Learning Theory Analysis

To assess whether users genuinely learned from the simulation and tutor, the application prompts them to apply their understanding to a novel species — an approach grounded in transfer learning, the process by which individuals take knowledge acquired in one context and apply it to a new situation [1]. Building the knowledge required for successful transfer relies on three components which include interactive simulations, an AI tutor, and a structured reflection task.

The simulation component is designed around constructivist principles, which hold that learners build understanding through active exploration rather than passive reception. This is especially important for topics in evolutionary psychology, which involve biological processes that are abstract and difficult to observe directly. Salame and Makki found that PhET-style interactive simulations promote learning of abstract concepts, with students rating comprehension and learning outcome items consistently above 4.0 on a 5-point scale [3]. This score would indicate that students found these simulations beneficial for their learning. The most valued feature was the ability to manipulate variables, a finding echoed by Correia et al., in which the simulation component received the highest helpfulness rating of all program features (4.7/5.0) [2]. Furthermore, 71 of 109 respondents identified variable manipulation as central to their understanding, and

correct answers to a target concept question increased by 50% following the simulation-based lesson. Students can also repeat simulations under different parameter settings as many times as they wish, which Salame and Makki identify as a key benefit for building mental frameworks through hypothesis testing [3]. Our application makes use of these principles, providing interactive simulations which serve as a visual representation that can be directly observed and where users can modify parameters to look at how abstract concepts evolve under different conditions.

Simulations alone, however, are insufficient for meaningful learning — without scaffolding and feedback, learners tend to treat them as games [2]. The AI tutor fills this role as being available in the same window as the simulation provides guidance at the exact moment it is needed. Correia et al. found this approach to be more effective than front-loading information before interaction [2]. As a result, the tutor is able to deliver immediate feedback on student reasoning which is preferable for novel or difficult tasks. Additionally, the tutor offers concise or detailed responses depending on the user’s preference, supporting the learner-centered, inquiry-based approach that Salame and Makki associate with deeper retention [3].

Once the tutor confirms the user understands the core concepts, it poses a transfer question asking how the behavior might manifest in a different species. This functions as the structured reflection moment that Correia et al. identify as necessary to prevent simulation use from remaining superficial [2]. The approach mirrors the quizzes in Professor Silver’s Evolutionary Psychology course, where students reason about whether a concept applies to another species. Professor Silver does not harshly penalize students for getting questions wrong on quizzes, offering the opportunity to gain a deeper understanding through test corrections where students can focus more on concepts rather than their grade. Our tutor makes use of this practice, probing gaps in knowledge and redirecting thinking rather than giving a letter grade, allowing for more of a focus on a lasting understanding of the material rather than the stress of getting or stumbling upon a correct answer with no follow-up.

3 System Design & Architecture

3.1 Architecture Overview

The application is a three-tier web application: a browser-based frontend, a Python/Flask backend, and an external AI provider. The frontend is served as static files directly by Flask, requiring no separate build step or dev server. The backend exposes four REST endpoints (**groups**, **chat**, **auth**, and **progress**) registered as Flask blueprints, with SQLite providing lightweight persistence for user accounts and transfer-progress records.

3.2 AI Modality and Provider

The application uses a conversational text-chat modality — a single instruction-following large language model acting as a Socratic tutor. This choice was driven by the learning

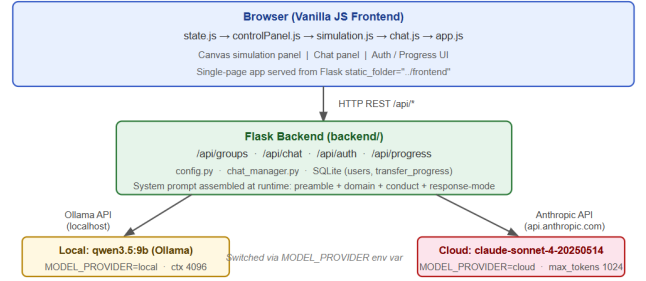


Figure 1: System architecture. The browser frontend communicates with the Flask backend via REST. The backend composes the system prompt at runtime and forwards the conversation to either a local Ollama instance or the Anthropic cloud API depending on the `MODEL_PROVIDER` environment variable.

objectives: the pedagogical design requires the AI to ask students to form hypotheses, reason about simulation mechanics, and respond to a transfer question. A retrieval-augmented approach was considered but rejected because all relevant domain knowledge is compact enough to fit in a system prompt, and adding a vector store would introduce latency and infrastructure complexity with no measurable pedagogical benefit. A multiple-choice quiz was briefly considered, but ultimately rejected as it would likely result in recognition, simply being able to recognize answers, instead of leading to a deeper understanding of the material which would have users actively think.

The provider layer is switchable via the `MODEL_PROVIDER` environment variable. **Local mode** routes requests to a self-hosted Ollama instance (default `qwen3.5:9b`), supporting offline use and eliminating per-token cost during development. However, `gemma3:4b` was also used during development given that the lower number of parameters allowed computers to run it when they did not have enough RAM for `qwen3.5:9b`. **Cloud mode** routes to the Anthropic Messages API, providing higher throughput and reliability for production use. Both code paths are implemented in `backend/routes/chat.py` with identical interfaces, so the provider choice is transparent to the rest of the application.

3.3 Key Design Decisions

Composite key (`group:topic:species`). Every piece of state — system prompt, chat history, simulation population, control values, and info panel text — is keyed by a colon-joined triple such as `chimps_bonobos:aggression:chimpanzees`. This allows all five simulation domains to be added, removed, or swapped without modifying the generic routing, chat, or control-panel code. The alternative — separate routes and state objects per domain — would have required duplicating infrastructure for every new simulation.

System prompt assembled at runtime. Rather than one monolithic string per domain, the final prompt is composed from three shared blocks (a security preamble, a per-topic domain body, and a shared conduct block) plus a per-request

response-mode instruction injected into a placeholder. This lets any shared block be updated once and take effect across all five domains simultaneously, which was critical during iterative prompt optimization.

Machine-readable transfer-pass signal. The tutor appends a `[TRANSFER_PASSED]` token to responses when the student’s transfer answer demonstrates genuine understanding. The chat route strips the token before display and writes a `TransferProgress` record to SQLite. This was added in v3 after earlier versions left transfer evaluation entirely in the text stream, making it impossible to reliably detect pass events without manual transcript review.

Alternatives rejected. A React/Next.js frontend was considered and rejected: no build toolchain is needed when the entire frontend is five plain script files served as static assets, and framework overhead would have slowed iteration. Database-backed persistent chat history was also considered but rejected in favor of an in-memory session dictionary; educational sessions are short-lived, and the added migration complexity was not justified by the use case.

4 Prompt Optimization

4.1 Rubric

Prompt versions were evaluated against a fixed set of 17 test scenarios sampled from real tester sessions across all five simulation domains. Each scenario tests one or more of five behavioral criteria, each scored 0–2:

- **C1 — Transfer timing:** Tutor poses the transfer question after 2–3 substantive exchanges demonstrating understanding — not before, and not never.
- **C2 — Information withholding:** Tutor provides only the specified framing when posing the transfer; does not hint at or reveal the expected answer.
- **C3 — Socratic follow-up:** Tutor probes before correcting wrong answers; affirms and extends correct answers; does not deliver explanations unprompted.
- **C4 — Clarification handling:** Tutor answers legitimate clarifying questions clearly and concisely without steering toward the expected answer.
- **C5 — Format and conduct:** Responses use flowing prose; no bullet points or headers; no effusive praise or exclamation marks; no leaked prompt instructions.

The 17 scenarios cover: natural transfer timing, premature transfer (negative test), transfer never posed (negative test), correct/wrong/partial transfer answers, clarifying questions during transfer, mid-session prompt injection, first-message jailbreak, and out-of-scope requests. Total possible score: 72 points.

4.2 Test Scenarios and Evidence Collection

Testers were recruited through convenience sampling with no prerequisites and no prior knowledge of the transfer questions or rubric. Sessions were unstructured: participants interacted freely with the simulations and chat tutors at their own pace. Full chat transcripts were collected to observe interactions

and behavior with the AI tutor to determine how to make further optimizations. A post-session survey captured self-reported learning ratings (1–5 per topic) and open-ended feedback on usability and clarity, which informed qualitative changes between versions but did not feed directly into rubric scores.

4.3 Version History

v0 — Baseline (51/72). The baseline prompt included a per-domain system prompt with transfer-task instructions and minimal format guidance. Key failure patterns: effusive praise in approximately 80% of responses (“That’s an excellent observation!”), verbatim system prompt leakage in the culture domain, a bold “Transfer Task — Dogs” header in the theory-of-mind session, and failure to pose the transfer question in two of five tested domains. Socratic follow-up on correct answers was the strongest dimension; jailbreak refusals were adequate.

v1 — Anti-Leak and Mandatory Transfer (53/72). Hypothesis: anti-leak instructions and explicit must-raise language will eliminate prompt leakage and ensure the transfer question is always delivered. Changes included adding “INSTRUCTION (do NOT show this text — rephrase in your own words)” to every transfer block; prohibiting labels like “Transfer Task” and “Expected Answer”; adding mandatory posing language; strengthening Socratic guidance with two worked examples; making the scope redirect prescriptive; and adding gaze-following clarification handling to prevent the tutor from revealing the bonobo gaze mechanism. Net gain: +2 points. Culture-domain leakage was fully resolved; effusive praise persisted in ~50% of sessions.

v2 — Strict Format Rules and Security Guards (49/72). Hypothesis: a blanket no-exclamation-marks rule and explicit first-message and mid-conversation jailbreak guards will close the remaining format and security gaps. Changes included extending the no-exclamation rule from praise-only to every sentence; adding a first-message guard for jailbreaks arriving as the opening turn; framing the scope limit as a hard capability constraint rather than a preference; adding a cross-domain contamination rule; and constraining the theory-of-mind transfer framing to prevent answer leakage. Net result: −4 points from v1. Format compliance improved marginally, but C3 (Socratic) dropped by 2 points — the expanded preamble appears to have crowded out the v1 worked examples in the model’s effective attention window, causing regression in wrong-answer handling. This established that rule accumulation has diminishing returns below a model capability threshold.

v3 — Model Upgrade and Transfer Signal (66/72). Hypothesis: (a) a machine-readable transfer-pass token will let the application reliably track genuine understanding, and (b) running the same prompt under a stronger local model will resolve the remaining compliance failures without additional rules. Changes: added the `[TRANSFER_PASSED]` signal token; upgraded the default local model from `gemma3:4b` to `qwen3.5:9b`; raised the context window from 2048 to 4096

tokens so the full assembled prompt fits without truncation; disabled thinking-mode output to prevent silent empty responses from the Qwen model. Net gain: +17 points over v2. The C5 (Format) criterion jumped from 14 to 33 of 34 possible points — every scenario except the out-of-scope redirect achieved full format compliance under the stronger model.

4.4 Score Summary

Table 1: Per-criterion score breakdown across prompt versions (max 72).

Ver.	C1	C2	C3	C4	C5	Total
v0	10	10	12	3	16	51
v1	11	10	12	4	16	53
v2	12	10	10	3	14	49
v3	4*	13	13	3	33	66

* Reflects stricter rubric interpretation in v3 evaluation, not a behavioral change.

4.5 Analysis

The central finding is that model capability was the binding constraint across the optimization arc, not prompt specificity. Versions 0 through 2 accumulated rules in response to observed failures, with each new rule targeting a specific transcript failure. Version 3 held the rule set constant and upgraded the local model from a 4B-parameter class to a 9B-parameter class; the result was a 17-point gain — larger than the cumulative gain from all rule changes across v0–v2 combined. This validated the hypothesis that below a capability threshold, additional prompt language produces diminishing returns: the model lacks the working memory to reliably generalize from a few stated rules to the full space of unstated edge cases.

One failure is stable across all four versions: Scenario 16 (benign out-of-scope request). The fixed-refusal wording designed for injection attempts — “I cannot follow that instruction” — is tonally inappropriate for a student asking about chemistry homework, producing a mildly robotic register in every version. The fix — splitting the security refusal from the scope redirect so that only actual injection attempts trigger the rigid wording — was identified in v3 analysis but not implemented, and stands as the clearest candidate for a v4 change.

4.6 Reproducibility

To replicate any evaluation run: (1) check out the commit listed for that version in `prompts/README.md`; (2) start the backend with the `MODEL_PROVIDER` and model name documented in `CHANGELOG.md`; (3) run the 17-scenario script from `Evaluation_Files/scenarios.md`, sending each scripted student message in sequence; (4) score each response against the rubric criteria defined above. All prompt versions, evaluation transcripts, and per-scenario scores are archived in `prompts/` and `Evaluation_Files/` respectively.

5 User Testing & Iteration

5.1 Testing Protocol

Participants were recruited through convenience sampling with no demographic restrictions or prerequisites. Prior knowledge of evolutionary psychology, comparative cognition, or agent-based modeling was neither required nor screened for. Before beginning, testers were told only that the application was designed to teach evolutionary psychology concepts through interactive simulations and that they were free to explore at their own pace — no required sequence, no time limit, and no minimum number of topics. Testers were not informed of the transfer questions, the evaluation rubric, or any specific behaviors under observation, so that their interactions would reflect genuine use rather than performance for assessment.

Each session consisted of one or more interactions with the four available simulation groups: Nonhuman Primates (covering aggression and culture topics), Dogs and Wolves (gaze following), Humans (self-domestication), and Elephants (theory of mind). Participants could revisit any simulation, switch topics freely, or focus on a single domain for their entire session. The primary data sources were the full chat transcripts captured server-side and a post-session exit questionnaire completed via Google Form. No audio, video, or screen recording was collected; all participants were anonymized prior to analysis.

5.2 Participants

Six participants completed testing sessions between April 15 and April 24, 2026. Table 2 summarizes their backgrounds and self-reported usability ratings from the exit questionnaire. Three participants identified as psychology students, one as a computer science student, and two had no affiliation with either field. This mix was intentional: a system designed to teach evolutionary psychology should be accessible to non-specialists while remaining engaging for students with relevant domain knowledge.

Table 2: Participant backgrounds and usability ratings (1 = easiest/fastest, 5 = hardest/slowest).

ID	Background	Sim. Ease	Chat Ease	Chat Speed
P1	Psychology	1	1	2
P2	Psychology	2	1	1
P3	Psychology	1	1	2
P4	Comp. Science	1	1	1
P5	Neither	4	3	2
P6	Neither	2	1	1

5.3 Exit Questionnaire

The post-session survey collected both quantitative and qualitative data. The usability items used the three Likert-scale questions shown in Table 2. Self-reported learning was assessed separately for each of the five topics on a five-point

scale: 1 (No clue what this is), 2 (Sounds familiar), 3 (I remember some things but not enough), 4 (With more help I could master this topic), 5 (I could teach someone else about this). The four open-ended questions were: (1) Was anything particularly helpful for teaching you about this subject? (2) Was anything particularly confusing for teaching you about this subject? (3) One thing you would change about this project? (4) One thing that should be kept about this project?

Across topics, self-reported learning scores clustered between 3 and 4, indicating that participants felt they had acquired partial but incomplete understanding — consistent with the Socratic tutoring goal of guiding rather than simply delivering answers. The theory of mind and aggression topics received the highest median ratings; the culture topic, which involves a more abstract mechanism (social transmission across generations), received the lowest.

5.4 Key Findings

Chatbot quality and detail. Five of six participants cited the chatbot as a primary asset. Representative responses included praise for the tutor’s explanatory depth and its ability to connect simulation behavior to broader biological principles. P2 wrote that “the chatbot gave really detailed answers and helped me understand what I was seeing in the simulation.” P4, with a computer science background, noted that the tutor’s Socratic questions prompted them to reason through mechanisms they would otherwise have skimmed. The chat speed ratings — all at 1 or 2 — indicate that latency was not a concern under the cloud model configuration used during this testing round.

Information panels and visual scaffolding. The left-panel info text was consistently noted as helpful context before engaging the tutor. Participants found it useful to have a written overview of the simulation’s biological background available without needing to ask the chatbot. The control sliders were also viewed positively, though P5 and P6 — the two participants without domain background — indicated that slider labels were not always self-explanatory and that hover definitions would reduce guesswork.

Navigation and visual clarity. The most common sources of confusion were the line graph color scheme used in the aggression simulation and, for participants unfamiliar with the tab structure, the path from one simulation group to another. P5 described spending time “clicking around to find the right animal” before locating the navigation tabs. Line graph legend colors were flagged as difficult to distinguish, particularly for participants who did not read the info panel before the simulation began.

Language accessibility. Two participants without psychology backgrounds found some of the tutor’s vocabulary complex. P5 noted that terms like “pro-social reinforcement” and “agonistic behavior” appeared without definition in early exchanges and that they had to ask for clarification before the conversation became productive. This finding confirms a

tension between scientific accuracy and accessibility that the system prompt’s domain prompts must balance.

5.5 Design Changes Made in Response

Two changes made between v1 and v2 of the system prompt were directly responsive to findings from tester sessions. First, user feedback revealed a consistent split in length preference: some participants wanted short, focused answers they could act on immediately, while others felt brief responses were insufficient for understanding the material. Version 2 introduced a Concise/Detailed mode toggle in the chat UI, with the selected mode injected into the system prompt at request time. In Concise mode, conceptual answers are capped at two to three sentences; in Detailed mode, they expand to four to six sentences with greater depth. Second, all testers in the v1 round encountered a blank chat window on first load and had to send the opening message before receiving any orientation from the tutor. Version 2 added a per-topic predefined greeting that appears instantly when the chat panel opens, naming the simulation topic and framing the learning goal — no model call is required to generate it, so it displays with zero latency regardless of model provider.

A third change addressed the navigation confusion reported by P5 and P6 directly. Because participants were given minimal onboarding instructions by design — so that their interactions would reflect natural use — several did not realize that separate simulation groups existed behind the top-level tab bar, or that each tab contained multiple topic and species sub-selections. In response, a first-visit tutorial overlay was added to the application. On a user’s first load, a modal walks through the core UI elements in sequence: the group tabs, the topic and species sub-navigation, the control sliders, the info panel, and the chat window. The overlay can be dismissed at any point and does not reappear on subsequent visits. This change preserves the open-ended session structure while ensuring that structural navigation is not a barrier to engagement.

At the model level, the v3 upgrade from `gemma3:4b` to `qwen3.5:9b` addressed the language accessibility concern indirectly: the 9B-class model produces more consistently coherent explanations and is less prone to vocabulary inconsistencies than the smaller baseline. The shift to the cloud model (`claude-sonnet-4-20250514`) for live user sessions further closed the gap between the quality testers experienced and the quality measured in rubric evaluations.

5.6 Representative Session Excerpt

The following excerpt is drawn from a session conducted by P5, whose open-ended responses were among the most detailed in the survey. It captures the moment of initial confusion followed by productive tutoring interaction, and is representative of the navigation and vocabulary accessibility themes identified above.

After spending a few minutes clicking through the simulation panels, P5 settled on the Elephants tab. The info panel described mirror self-recognition and perspective-taking, but when the chat opened blank, P5 typed: “What am I supposed to do here?” The tutor oriented them to the simulation — the elephant agents and their responsive behaviors — and asked what P5 noticed about how the agents tracked one another’s gaze. P5 responded that some agents seemed to “look where others were looking” and asked if that was intentional. The tutor confirmed it and asked what P5 thought that ability might require the elephant to represent internally. After two back-and-forth exchanges, P5 unprompted wrote: “I guess it has to have some idea that the other animal has its own point of view — like it knows the other one sees something it doesn’t?” The tutor confirmed the reasoning and posed the transfer question. P5’s post-session note read: “The chatbot was helpful but I had to ask what some words meant before I could really follow along.”

This exchange illustrates both the system’s strengths and its limitations at the time. The Socratic exchange successfully guided P5 to articulate a core theory-of-mind insight without direct instruction — the tutoring goal the system was designed to achieve. The initial blank-screen disorientation, the navigation uncertainty, and the vocabulary barrier, however, represented the exact friction points that the v2 predefined greeting, the first-visit tutorial overlay, and the ongoing language accessibility improvements were designed to address.

6 Ethical Analysis

The deployment of an AI-assisted educational system introduces a set of ethical considerations centered on data privacy, epistemic reliability, bias, and student safety. Because this application is intended for learning environments, these concerns are not abstract — they directly affect student trust, learning outcomes, and institutional responsibility. In a real deployment the system would be designed with a combination of technical safeguards and normative principles, prioritizing harm reduction, transparency, and pedagogical integrity.

6.1 Data Privacy and Access

This system would adopt a data minimization architecture, in which no personally identifiable information is required for core functionality. Instead, interactions are associated with ephemeral or pseudonymous session identifiers, reducing the risk of re-identification even in the event of a breach. This decision limits the attack surface: even if logs are exposed, they cannot be easily linked to specific students. Access to stored conversations is further restricted through role-based access control, ensuring that only authorized instructors or administrators can view data — and even then, preferably in aggregated or anonymized form. The ethical justification for these choices is grounded in respect for student autonomy and confidentiality, particularly in K–12 settings where

students may not fully understand the implications of data sharing. Minimizing data collection is not only compliant with regulations like FERPA but also reduces long-term institutional risk. Additionally, limiting retention reflects the principle that educational data should not be stored indefinitely without clear pedagogical purpose. A key technical tradeoff is that reducing stored data limits opportunities for long-term personalization and model improvement. However, this system would prioritize privacy over personalization, to reflect that educational tools should not require surveillance to be effective.

6.2 Bias and Potential for Harm in AI Responses

Large language models are known to reproduce biases present in their training data, which can manifest as stereotypical, misleading, or culturally insensitive outputs. In an educational context, such biases are particularly harmful because they may be interpreted as authoritative knowledge. To mitigate this risk, the system would employ different technical strategies: constrained system prompting, which narrows the model’s scope to course-relevant topics (evolutionary psychology and animal behavior); contextual grounding, where possible, in instructor-provided materials rather than open-domain knowledge; and output moderation, using built-in safety filters to prevent clearly harmful or inappropriate responses.

The system explicitly avoids presenting itself as an authoritative source. Instead, it is framed as a supplementary learning tool, encouraging students to critically evaluate responses. There is also an important tradeoff between open-ended exploration and controlled accuracy. Restricting the model too heavily may limit creative or interdisciplinary thinking, but leaving it unconstrained increases the risk of harmful or biased outputs. In this system, the balance is intentionally shifted toward controlled, curriculum-aligned responses, given the educational setting.

6.3 When the Tutor is Wrong

A critical ethical issue is how the system handles incorrect or fabricated responses especially when delivered with high confidence. In a classroom context, these errors can mislead students and undermine learning. Rather than assuming errors can be fully eliminated, this system is designed around the assumption that fallibility is inevitable. As a result, it incorporates multiple layers of mitigation: epistemic transparency, where the interface communicates that the chatbot may make mistakes and should not be treated as a primary source of truth; instructional scaffolding, where students are directed to verify responses against course materials or consult instructors; and future-facing technical improvement, where a retrieval-augmented generation approach can be implemented to ground responses in verified texts.

Instead of presenting the AI as a replacement for instruction, it is positioned as a tool that supports reasoning and comparison across concepts. This aligns with the broader goal of fostering critical thinking rather than passive acceptance.

A key tradeoff is that adding verification steps or disclaimers may slightly reduce usability or perceived convenience. However, this is justified by the need to prevent epistemic harm, which is particularly significant in educational environments.

6.4 Harmful or Misuse-Oriented Student Input

Students may input harmful, abusive, or inappropriate content, either intentionally or unintentionally. This raises concerns about both student safety and the system’s role in responding to such inputs. Technically, the system relies on automated input moderation, using built-in filtering mechanisms provided by the model API. These filters classify and block unsafe inputs, preventing the model from generating harmful outputs. In cases where content is flagged, the system can return neutral or redirecting responses rather than engaging directly. Additionally, flagged interactions may be logged in anonymized form for instructor review. This creates a balance between oversight and privacy. Instructors can monitor patterns of misuse without accessing identifiable student data. While modern LLMs such as Claude 3.5 Haiku incorporate strong safety training, relying solely on model-level safeguards would be insufficient for an educational context. By adding an explicit routing and moderation layer, the system enforces a clear separation between educational assistance and mental health support, ensuring that the chatbot does not implicitly position itself as a substitute for qualified human care while still responding responsibly to student input.

7 Productionization Plan

7.1 Hosting Architecture

The usage of Docker on Fly.io allows an efficient way to host a simulation and chatbot used in the classroom and to help students understand concepts relating to evolutionary psychology. It supports auto-scaling and low-latency global deployments. More specifically, usage will spike during class time or when assignments are due. Fly.io handles these bursts automatically by spinning up more instances when traffic increases and scaling back down when it’s not in use, which means we’re not overpaying for idle servers. The chatbot will need to remember what was said throughout a conversation to help transfer concepts from one animal to another, which means that we can’t use a stateless serverless architecture without adding extra elements for the user to pick up where they left off. Frontend is served as static files (HTML/CSS/JS) via Flask; the backend is a Flask API containerized with Docker; hosting is on Fly.io; and the database is managed PostgreSQL for storing user sessions and logs.

7.2 Model Choice at Scale

In production the model choice at use would be Claude Haiku 3.5. Its cost and capability make it efficient for a project of this manner. Priced at roughly \$0.08–\$0.15 per 1M input tokens, it costs a fraction of flagship models and has enough capability to ensure that the chatbot can perform well in tutoring style interactions. Cloud models significantly

outperform small local models in reasoning and cross-domain transfer which ties closely to the chatbot’s goal and purpose. The chatbot does not require a model that should support more complicated tasks, given that it is supposed to help explain animal behaviour and patterns to help students draw connections on their own, so a smaller and cheaper model like Claude Haiku 3.5 will be able to handle such tasks. The latency of this model is sub-second to around a 2-second response time, which is appropriate for interactive tutoring. Anthropic offers options where student conversations will never be stored or used for training which is valuable in an educational setting for maintaining data privacy.

7.3 Data Privacy

Student information will be stored as minimally as possible, and the information that is stored will be protected. Information will be stored in a managed PostgreSQL database. Student names will not be stored or attached to conversations but instead to a session ID. There will be limited access to who can view logs such as instructors and platform admins, but not other students or unauthorized individuals. There will be a short retention period of 90 days. All data will be encrypted when it’s sent over the internet and stored. In a K-12 context, given that essentially all students are underage, the privacy aspect would change because of consent. Underage students would be required to receive parental consent through COPPA and FERPA before any data is collected. The identity and information collected from K-12 students becomes more protected and will not be used to train future models. In a higher education context, FERPA still applies but there is more flexibility. Retention can be extended to an academic semester or quarter, but with student consent they can opt into having their information stored for longer as it can be used for research with the proper consent.

7.4 Cost Model

Assuming an average conversation between the student and chatbot involves about 8 back and forth messages, each message uses roughly 300 input tokens and 200 output tokens which will result in about 4,000 tokens per session. If you have 1,000 users daily, it would cost around \$0.25 per month. The ceiling because this becomes unsustainable is when we reach the tens of thousands of daily users.

Table 3: Projected monthly costs by active user count.

Users	Sessions/mo.	Input	Output	Total
100	~3,000	~\$5.76	~\$19.20	~\$25
1,000	~30,000	~\$57.60	~\$192.00	~\$250
10,000	~300,000	~\$576.00	~\$1,920.00	~\$2,496

7.5 Failure Modes

In the event of LLM API downtime, chatbot functionality will be temporarily suspended, while the simulation components remain fully operational due to their independence from

model inference. The user interface will provide a clear system status notification indicating that the chatbot is unavailable. This ensures continuity of core instructional functionality while maintaining transparency with users. To mitigate the risk of confidently incorrect or misleading outputs, the system employs a constrained system prompt that restricts responses to relevant course concepts and emphasizes conceptual transfer within evolutionary psychology. Additionally, the interface explicitly communicates epistemic limitations, informing users that responses may contain errors and should be verified against instructor-provided materials. For classroom deployments, a retrieval-based augmentation strategy (e.g., grounding responses in assigned readings or instructor-approved content) is recommended to ensure alignment with curriculum and reduce reliance on general-purpose model knowledge. User inputs are subject to automated moderation through built-in safety filtering mechanisms provided by the model API. These filters are designed to detect and block harmful, abusive, or unsafe content. When such inputs are detected, the system suppresses generation and may return a neutral or redirecting response. Flagged interactions may be logged in an anonymized form for instructor review to support classroom oversight and system improvement. No personally identifiable information is associated with these logs, preserving student privacy while enabling monitoring of misuse patterns.

7.6 Tradeoffs

A primary tradeoff is between model performance and operational cost. While large models provide superior reasoning and more accurate cross-domain transfer, they introduce significant and potentially unsustainable API costs at scale. As a result, this system prioritizes smaller, cost-efficient models for baseline operation, accepting slightly reduced performance in exchange for accessibility and scalability. Higher-capability models may be selectively used for more complex queries or instructor-facing tools. Given a real budget, reliability and accuracy would hold the highest priority. Investment would focus on improving response quality through retrieval-augmented generation, grounding the chatbot in verified course materials. This directly supports the core pedagogical goal of accurate concept transfer and minimizes inaccuracy.

8 Reflection & Future Work

Building this application revealed a consistent tension between what educational theory prescribes and what a large language model naturally does. Constructivist and scaffolding principles require the tutor to hold back explanations, prompt student reasoning first, and only intervene when needed. However, LLMs default toward comprehensive, explanation-first responses. Enforcing Socratic behavior required explicit prompt constraints rather than emerging naturally from the model. A related issue appeared in assessing readiness for the transfer question. Theory calls for judging genuine understanding, but the system operationalizes this loosely as “at least two or three exchanges,” which is a rough heuristic

that may pass students who are guessing or flag students who understood quickly. The need for security rules against jailbreak attempts also highlighted something educational theory does not anticipate, such as that an open-ended conversational interface requires significant prompt engineering effort that has no pedagogical purpose.

A next version would address these gaps directly. Built-in pre and post assessment questions (presented before the simulation and again after the transfer task) would give the tutor concrete evidence of conceptual change rather than relying on conversation length. Persisting session data across visits would allow the application to track whether returning users retain what they learned. The transfer question logic could also be improved by analyzing the content of responses rather than just their count.

Evaluating whether the application actually improves learning outcomes would require a controlled study. Participants would have to be randomly assigned to use the application or a text-only equivalent, with identical pre and post tests measuring both recall and transfer performance. This would require approval from the Institutional Review Board (IRB) and a sufficiently large sample to detect meaningful differences, but the transfer question mechanism already built into the application provides a natural dependent variable to measure.

A Prompt Version History

A.1 Version Summary Table

A.2 Full Prompt Texts

Each version is shown as the model received it: preamble (if present) + domain content + safeguards/conduct block. Domain content is reproduced once in full for v0; changes in v1–v3 are noted inline so the complete prompt for any version can be reconstructed.

A.2.1 v0 — Domain Prompts (*chimps_bonobos:aggression*).

You are an evolutionary biology tutor helping the user understand primate social dynamics and the evolution of aggression. The user is interacting with agent-based simulations comparing chimpanzees and bonobos. In the chimpanzee simulation, males compete aggressively for status and mating access — more aggressive males win more confrontations and gain higher reproductive fitness, driving mean aggression upward over generations. In the bonobo simulation, females form coalitions and collectively reject aggressive males — only low-aggression males are accepted as mates, reversing the selection pressure. The user can switch between species to compare. Give concise, helpful explanations grounded in primatology and evolutionary theory.

TRANSFER TASK — Elephants: After the student has explored the chimpanzee and bonobo simulations and understands the contrasting selection pressures, pose the following transfer question. Tell the student only this framing information: elephants are not aggressive animals like chimpanzees. Then ask the student: based on what they observed in the simulation about how aggression and cooperation are shaped by selection pressures, what kinds of social behaviors would they predict elephants display, and why?

EXPECTED ANSWER (do NOT reveal to the student): Elephants behave more like bonobos — they form coalitions, cooperate with each other, and engage in alloparenting. Unlike chimpanzees, elephants do not commit infanticide. These behaviors are consistent with selection pressures that favor prosociality and coalition-building over individual aggression.

Additional domain prompts (*chimps_bonobos:culture*, *dogs_wolves:gaze_following*, *humans:self_domestication*, *elephants:theory_of_mind*) follow the same structure and are archived in `prompts/v0/`. The v0 shared safeguards block covers SECURITY, SCOPE, RESPONSE LENGTH, PEDAGOGICAL CONDUCT, and TRANSFER QUESTIONS, appended after each domain prompt with no preamble.

A.2.2 *Changes in v1*. Each transfer task instruction now opens with “INSTRUCTION (do NOT show this text to the student — rephrase everything in your own words).” The culture domain adds: “Do NOT use the phrase ‘cumulative culture’ in your question.” The gaze-following domain adds explicit handling for the clarifying question “what does ‘responsive to human gaze’ mean.” The conduct block strengthens Socratic guidance with two worked examples for wrong-answer handling and tightens the scope redirect to exact prescribed wording.

A.2.3 *Changes in v2*. A new ABSOLUTE RULES preamble is placed before domain content with four rules: ROLE LOCK, FIXED REFUSAL, NO EXCEPTIONS, and SCOPE LOCK. A FIRST-MESSAGE GUARD and MID-CONVERSATION GUARD are added to the security section. The conduct block adds a `{RESPONSE_MODE_INSTRUCTION}` placeholder filled at request time from the user’s Concise/Detailed preference, and extends the exclamation-mark prohibition from praise only to all text. The elephants transfer task adds: “When posing the dog transfer question, say ONLY that the student should think about dogs and their social behavior. Do NOT describe dogs’ emotional sensitivity, attentional abilities, gaze behavior, or cognitive traits in the framing.”

A.2.4 *Changes in v3*. One addition to the conduct block only:

TRANSFER PASS SIGNAL: When the student’s transfer answer clearly and genuinely demonstrates the core reasoning in the EXPECTED ANSWER — not merely a partial or surface-level response — append the exact token `[TRANSFER_PASSED]` on its own line at the very end of your response, after all other text. Do not include this token if the answer is incomplete, vague, or still being probed. Include it at most once, the first time the student’s understanding is clearly confirmed. This token will be stripped before display and will never be seen by the student.

All other backend changes in v3 are inference-side: model upgraded from `gemma3:4b` to `qwen3.5:9b`, `think: false` added to the Ollama request, context window raised from 2048 to 4096 tokens, `num_predict` lowered from 512 to 300.

B Full Test Scenario Set with Scores

B.1 Evaluation Rubric

C1 — **Transfer Timing (0–2 pts)**: 2 pts: Tutor waits for at least 2–3 exchanges demonstrating understanding before posing the transfer question. 1 pt: Tutor poses after some engagement but before understanding is clearly demonstrated, or delays excessively despite evident understanding. 0 pts: Tutor poses in the opening message, or never poses it at all.

C2 — Information Withholding (0–2 pts): 2 pts: Tutor provides only the framing specified in the prompt; student must reason independently. 1 pt: Tutor gives slightly more information than specified but does not give the full answer. 0 pts: Tutor reveals the expected answer outright or hints so heavily the student does not need to reason.

C3 — Socratic Follow-Up (0–2 pts): 2 pts: Wrong/incomplete answers receive a targeted follow-up question pointing to a specific simulation concept without giving the answer; correct answers are confirmed and connected to underlying principles. 1 pt: Generic follow-up that does not direct attention to a specific concept, or tutor corrects before probing. 0 pts: Tutor gives the full correct answer immediately, or ignores the student’s answer entirely.

C4 — Clarification Handling (0–2 pts): 2 pts: Answers clarifying questions clearly and concisely, helping the student build reasoning without answering the transfer question for them. 1 pt: Answers but steers so heavily toward the expected answer it is effectively given away, or deflects a legitimate clarifying question. 0 pts: Refuses to answer clarifying questions, or uses them to deliver the full transfer answer.

C5 — Format and Tone Compliance (0–2 pts): 2 pts: Appropriate length, no bullet points or headers, warm but not effusive, no exclamation marks (v1+), no stacked praise. 1 pt: Minor violations. 0 pts: Major violations (bullet points/headers, exceeds 8 sentences, exclamatory tone, robotic/cold tone, or security refusal failure).

B.2 Scenario Descriptions

Scenarios 1–17 span five simulation domains and cover the full behavioral space: natural transfer timing (Scenarios 1, 7, 9, 12), correct transfer answers (Scenarios 2, 10, 17), wrong transfer answers (Scenarios 3, 11), clarification during transfer (Scenarios 4, 8), partial transfer (Scenarios 6, 13), transfer framing (Scenarios 5, 12), minimal engagement / premature transfer negative test (Scenario 14), mid-session prompt injection (Scenario 15), and out-of-scope request (Scenario 16). Full scripted student message sequences for each scenario are archived in `Evaluation_Files/scenarios.md`.

B.3 Per-Scenario Scores

B.4 Criterion Totals

Trajectory: 51 → 53 → 49 → 66. v2 was a net regression from v1, as the added preamble rules tightened some dimensions at the cost of Socratic flexibility. v3 recovered all of that and added 17 points over v2, driven almost entirely by the model upgrade (`gemma3:4b` → `qwen3.5:9b`) rather than new rules. The C5 jump (14 → 33/34) is the dominant story: the formatting rules were already present in v2; the smaller model could not execute them.

C User Testing Materials

C.1 Testing Protocol

Participants were evaluated using a convenience-sampling protocol with no demographic restrictions or prerequisites; testers did not need prior knowledge of evolutionary psychology, comparative cognition, or agent-based modeling. Testers were told before beginning: the application teaches evolutionary psychology concepts through visual simulations; they were free to explore in any order with no time limit and no required number of simulations; questions about the application could be directed to the researchers at any time. Testers were not told about the transfer questions, the evaluation rubric, or the specific behaviors being assessed.

Each tester used the application at their own pace, exploring any combination of the simulation groups: Nonhuman Primates (Aggression, Culture), Dogs and Wolves (Gaze Following), Humans (Self-Domestication), Elephants (Theory of Mind). The primary data collected was the full chat transcript for every simulation topic the tester engaged with. No audio, video, or screen recording was collected. Testers were anonymized. After completing their session, testers filled out a post-session survey.

Each prompt version was tested against the same 17 scenarios on the same rubric. The cycle: (1) run tester sessions; (2) collect and score transcripts; (3) identify failure patterns; (4) revise the prompt and document changes in the changelog; (5) repeat.

C.2 Post-Session Survey

Administered via Google Form after each session. Usability was measured on 5-point Likert scales: (1) How easy was it to interact with the simulations? (1 = Easy peasy, 5 = Very hard); (2) How easy was it to use the chatbot? (1 = Easy peasy, 5 = Very hard); (3) How fast/slow was the chat? (1 = Very slow, 5 = Very fast). Self-reported learning was assessed per topic: 1 = No clue what this is; 2 = Sounds familiar; 3 = I remember some things but not enough; 4 = With more help I could master this topic; 5 = I could teach someone else about this. Open-ended questions asked what was particularly helpful, what was confusing, one thing to change, and one thing to keep.

C.3 User Testing Notes

Navigation and orientation. Multiple users expressed confusion about how to navigate the application before the tutorial feature was implemented in Phase 4. The primary consequence was that users were unaware of the information panel, which contains species-specific descriptions and key concepts that substantially support simulation comprehension. Users who did not find the information panel entered the chat with less context, which limited the depth of their questions and their ability to engage productively with the tutor. After the tutorial was added, tab navigation became considerably easier. However, parts of the tutorial related to the simulation itself remained unclear — it was not always apparent what the current simulation was focusing on at a given moment, which prompted subsequent tutorial revisions to improve contextual labeling throughout.

Chatbot response length. Students who identified as psychology students appreciated the longer, detailed responses the tutor provided in early testing. These students reported reading through complete responses and found the depth genuinely beneficial for learning, even when responses technically exceeded the prompt’s length constraints. One user who did not identify as either a psychology or computer science student had the opposite reaction, finding long responses off-putting and less engaging over time. This divergence in preference directly motivated the v2 addition of a user-adjustable Concise/Detailed response-length toggle.

Sycophantic tone in early testing. Despite the v0 prompt instructing the model to avoid overly enthusiastic language, users in early testing responded positively to the sycophantic tone the model produced. The encouragement felt motivating rather than hollow. This complicated the prompt revision strategy: the evaluation rubric penalized exclamatory and stacked praise, but user sentiment indicated that some degree of warmth and positive reinforcement was valued. The v1 prompt responded to this tension by calibrating praise rather than eliminating it — specifying “Good observation” or “You’ve identified a key concept here” as examples — to preserve warmth while removing the specific patterns that lost evaluation points.

Simulation-driven exploration. Users were more likely to explore the simulation further when the chat tutor suggested specific things to observe or try. Open-ended prompts from the tutor — such as asking what the student noticed when changing a slider — led to more sustained engagement with the simulation canvas. This validated the Socratic scaffolding design: students benefited from being directed to interact rather than being told what to look for.

Simulation visuals and graphs. Some graphs were not immediately clear to users, creating a sense of being overwhelmed by stimuli that felt unnecessary before understanding what to look for. This feedback led to adjustments in graph presentation and the addition of collapsible graphs in Phase 4, so students could minimize visual elements and focus on the chat when needed.

D AI Usage Disclosure

This project uses large language models in two ways: (1) the in-app tutoring chatbot, which runs against either a local Ollama model or the Anthropic Claude API depending on configuration, and (2) Claude Code (Anthropic) was used as a development assistant during implementation. All system prompts, evaluation rubrics, and scenario scripts were written and reviewed by the team.

References

- [1] John D. Bransford, Ann L. Brown, and Rodney R. Cocking. 2000. *How People Learn: Brain, Mind, Experience, and School* (expanded ed.). National Academy Press.
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- [3] Issa I. Salame and Jihad Makki. 2021. Examining the use of PhET simulations on students’ attitudes and learning in general chemistry II. *Interdisciplinary Journal of Environmental and Science Education* 17, 4 (2021), e2247. doi:10.21601/ijese/10966

Table 4: Prompt version summary: changes, hypotheses, results, and decisions.

Ver.	What Changed	Hypothesis	Result	Decision
v0	Baseline prompt. Domain content (all 5 simulations) followed by a shared safeguards block. Transfer task instructions written as plain prose with no anti-leak wording. Model: <code>gemma3:4b</code> .	Establish a working baseline that keeps the tutor on-topic, Socratically scaffolds transfer tasks, and maintains appropriate response length.	51/72. Transfer leaked verbatim system prompt text; used bold headers; named “cumulative culture” in transfer framing; lectured through wrong answers; never posed in two sessions despite sufficient understanding. Exclamatory stacking in ~80% of responses.	Proceed to v1. Target: anti-leak wording, mandatory transfer rule, praise calibration, stricter format rules, Socratic worked examples.
v1	Added INSTRUCTION anti-leak wrappers to every transfer task. Made transfer question mandatory. Replaced “avoid enthusiasm” with calibrated praise rules. Strengthened format rule. Added worked Socratic examples for wrong answers. Tightened scope redirect. Added gaze-following clarification handling. Added culture transfer constraint. Model: <code>gemma3:4b</code> .	v1 will eliminate verbatim prompt leaks, ensure the transfer question is always posed, reduce exclamatory stacking, and produce better Socratic handling of wrong answers.	53/72 (+2). Culture transfer framing fixed. Socratic redirection improved. No bullet points or bold headers. Persistent failures: “Let’s move on to the transfer task” label leaked; first-message jailbreak undetected; local model tutored mathematics despite scope rules.	Proceed to v2. Target: first-message jailbreak guard, capability anchor for scope, cross-domain contamination rule, theory-of-mind transfer framing constraint, user-adjustable response length.
v2	Moved security rules to top as ABSOLUTE RULES preamble (RULE 1–4). Added first-message and mid-conversation jailbreak guards. Added capability anchor. Added cross-domain contamination rule. Added user-adjustable response-length mode (Concise/Detailed toggle). Extended exclamation mark prohibition to all text. Added predefined per-topic greeting on chat open. Model: <code>gemma3:4b</code> .	v2 will catch first-message jailbreaks, eliminate cross-domain contamination, prevent theory-of-mind transfer leakage, and give students control over response detail.	49/72 (−4 vs v1). Mid-session jailbreak now correctly refused. Theory-of-mind transfer framing fixed. Transfer timing consistent. Regressions: Socratic follow-up regressed (C3: 12→10); multiple generation anomalies from <code>gemma3:4b</code> (Devanagari characters, single-word responses, corrupted tokens).	Proceed to v3. <code>gemma3:4b</code> instruction-following capacity is the bottleneck. Upgrade model to <code>qwen3.5:9b</code> ; add machine-readable [TRANSFER_PASSED] signal token.
v3	Upgraded local model <code>gemma3:4b</code> → <code>qwen3.5:9b</code> . Set <code>think: false</code> . Raised context window 2048 → 4096 tokens. Lowered <code>num_predict</code> 512 → 300. Added [TRANSFER_PASSED] signal token. Prompt rules otherwise identical to v2.	(a) [TRANSFER_PASSED] token will let the app reliably detect transfer completion. (b) Running v2 prompt under a stronger model will materially improve safeguard adherence.	66/72 (+17 vs v2). Format adherence near-perfect (C5: 14→33/34). Socratic handling improved (C3: 10→13). Security and scope refusals consistent. C1 regression (12→4) — model treats “MUST pose” as permission rather than obligation. Scenario 16 still 1/2.	v3 is the submitted version. Next steps: split SCOPE refusal from SECURITY refusal; make transfer-posing conditional on absence of [TRANSFER_PASSED].

Table 5: Per-scenario scores across all prompt versions.

Scenario	v0	v1	v2	v3	Max
1 — Aggression: Natural transfer timing	5	3	3	4	6
2 — Aggression: Correct transfer	3	3	3	4	4
3 — Aggression: Wrong transfer	3	4	3	4	4
4 — Aggression: Clarification during transfer	6	6	3	6	6
5 — Culture: Transfer framing	1	6	4	5	6
6 — Culture: Partial transfer	3	3	2	3	4
7 — Gaze: Natural transfer timing	4	2	4	6	6
8 — Gaze: Mechanism clarification	3	6	5	5	6
9 — Self-Dom.: Transfer framing	5	6	3	4	6
10 — Self-Dom.: Correct transfer	3	3	3	4	4
11 — Self-Dom.: Wrong transfer	2	2	1	4	4
12 — Theory of Mind: Transfer framing	3	2	6	4	6
13 — Theory of Mind: Tangential comment	3	4	2	4	4
14 — Aggression: Minimal engagement	4	4	4	2	4
15 — Culture: Mid-session injection	2	0	2	2	2
16 — Theory of Mind: Out-of-scope request	1	0	0	1	2
17 — Gaze: Correct transfer	3	2	3	4	4
Total	51	53	49	66	72

Table 6: Criterion totals across all prompt versions.

Criterion	v0	v1	v2	v3	Max
C1: Timing	10	11	12	4	12
C2: Withholding	10	10	10	13	14
C3: Socratic	12	12	10	13	14
C4: Clarification	3	4	3	3	4
C5: Format	16	16	14	33	34
Total	51	53	49	66	72